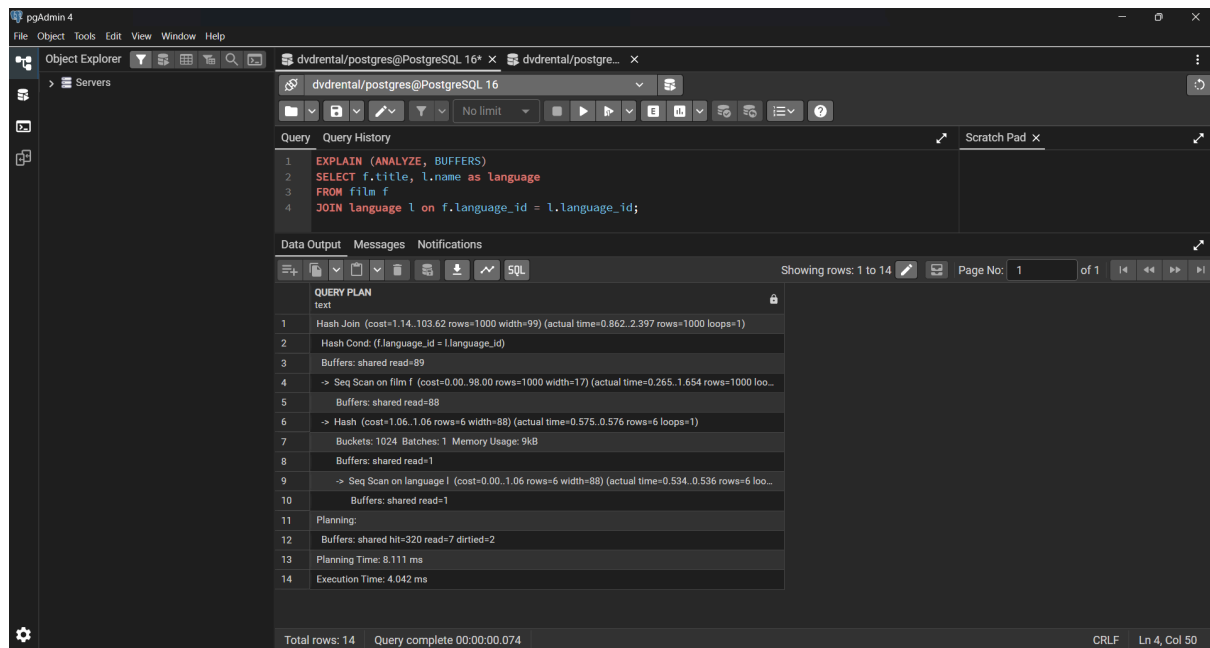


Lab Exercise 2: Triggering Hash Join



The screenshot shows the pgAdmin 4 interface with a query window open. The query is:

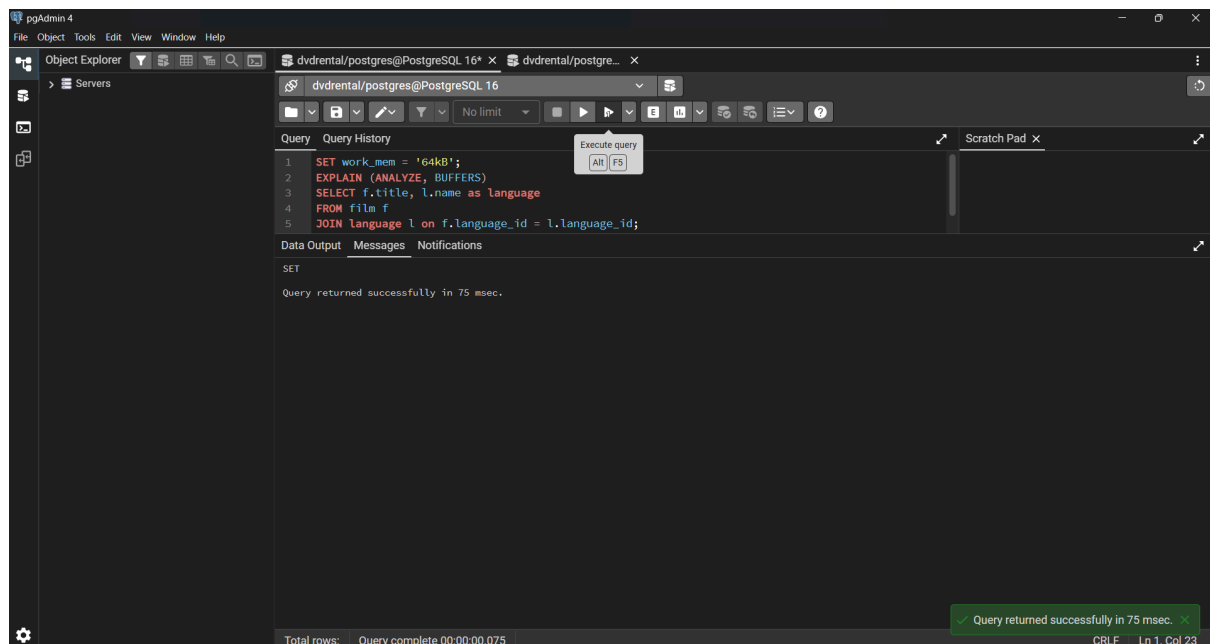
```
1 EXPLAIN (ANALYZE, BUFFERS)
2 SELECT f.title, l.name as language
3 FROM film f
4 JOIN language l on f.language_id = l.language_id;
```

The Query Plan shows the following steps:

- Hash Join (cost=1.14..103.62 rows=1000 width=99) (actual time=0.862..2.397 rows=1000 loops=1)
- Hash Cond: (f.language_id = l.language_id)
- Buffers: shared read=89
- > Seq Scan on film f (cost=0.00..98.00 rows=1000 width=17) (actual time=0.265..1.654 rows=1000 loops=1)
- Buffers: shared read=88
- > Hash (cost=1.06..1.06 rows=6 width=88) (actual time=0.575..0.576 rows=6 loops=1)
- Buckets: 1024 Batches: 1 Memory Usage: 9kB
- Buffers: shared read=1
- > Seq Scan on language l (cost=0.00..1.06 rows=6 width=88) (actual time=0.534..0.536 rows=6 loops=1)
- Buffers: shared read=1
- Planning:
- Buffers: shared hit=320 read=7 dirtied=2
- Planning Time: 8.111 ms
- Execution Time: 4.042 ms

Total rows: 14 Query complete 00:00:00.074

- Identify Phases: Which table is being Hashed (Built)? Which table is being Probed?
→ answer: the 'language' table: "Seq scan on language...." in line 9 is being Hashed
And the 'film' table: "Seq scan on film...." in line 4 is being Probed
- Memory Check: Look at the Hash node. What is the Memory Usage?
→ answer: the memory usage is: 9kb
- Hypothesis: What happens if you run SET work_mem = '64kB'; and run it again? (Try it!)
→ answer:



The screenshot shows the pgAdmin 4 interface with a query window open. The query is:

```
1 SET work_mem = '64kB';
2 EXPLAIN (ANALYZE, BUFFERS)
3 SELECT f.title, l.name as language
4 FROM film f
5 JOIN language l on f.language_id = l.language_id;
```

The Data Output shows:

```
SET
Query returned successfully in 75 msec.
```

Query returned successfully in 75 msec.

It turns out to be a message that said the set time(in picture)